

Manufacturer
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Digital Material Passport

<i>ID</i>	DMP-TEST-MULTIROW-001	<i>Version</i>	1.0.0
<i>Issue Date</i>	2025-04-10	<i>Certificate Type</i>	EN 10204 3.1

Business Transaction

Order		Delivery	
<i>Order ID</i>	ORD-2025-3312	<i>Delivery ID</i>	DLV-2025-4455
<i>Quantity</i>	18000 kg	<i>Quantity</i>	18000 kg

Product Information

<i>Product Name</i>	Forged round bars
<i>Batch ID</i>	BATCH-2025-03-7832
<i>Surface Condition</i>	Rough turned
<i>Delivery Condition</i>	+N - (Normalized)

Product Norms

<i>Standard</i>	EN 10250-1
<i>Standard</i>	EN 10250-2

Material Designations

<i>Name (EN 10027-1)</i>	S355J2 +N
<i>Number (EN 10027-2)</i>	1.0577 +N

Product Shape

<i>Form</i>	RoundBar
<i>Length</i>	2800 mm
<i>Diameter</i>	350 mm

Packaging and Marking

<i>Marking</i>	Both ends stamped with grade and heat number.
<i>Packaging</i>	Bundled, max 20 MT
<i>Coloring</i>	blue

Heat Treatment

Process	Lot	Furnace	Date	
Normalizing	HT-LOT-2025-0315-B	FURNACE-07	2025-03-15	
Stages				
Stage	Temperature	Duration	Cooling	Atmosphere
Austenitizing	910 C			Air

Chemical Analysis

Heat Number	H-2025-7832
Melting Process	EAF+LF+VD
Casting Method	IngotCasting

Elements

Symbol	C	Si	Mn	P	S	Cr	Ni	Mo	Cu
Unit	%	%	%	%	%	%	%	%	%
Min	0.1	-	1.1	-	0.015	-	-	-	-
Max	0.22	0.55	1.65	0.03	0.03	0.35	0.35	0.1	0.3
Actual	0.17	0.28	1.38	0.014	0.022	0.18	0.14	0.05	0.22
Symbol	B	Al	V	Sn	Ti	Sb	Ti	Sb	F1
Unit	%	%	%	%	%	%	ppm	%	%
Min	-	0.015	-	-	-	-	-	-	-
Max	-	0.06	0.025	0.015	0.006	0.003	3.0	0.5	0.52
Actual	4.0E-4	0.028	0.012	0.006	0.004	0.0012	2.1	0.38	0.37

Formula Definitions

F1 = Cr+Ni+Mo: 0.37%

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength in delivery condition Specimen: 1/2T, L - ID: TS-001	Rm	548N/mm²	440N/mm²	630N/mm²	EN ISO 6892-1/B1	-
Yield Strength in delivery condition Specimen: 1/2T, L - ID: YS-001	Re	372N/mm²	320N/mm²		EN ISO 6892-1	-
Elongation in delivery condition Specimen: 1/2T, L - ID: EL-L-001	A	26%	17%		EN ISO 6892-1/B1	-
Elongation in delivery condition Specimen: 1/2T, T - ID: EL-T-001	A	20%	14%		EN ISO 6892-1/B1	-
Reduction of area in delivery condition Specimen: 1/2T, L - ID: RA-001	Z	48%			EN ISO 6892-1	-
Impact test ISO-V -20°C, in delivery condition Specimen: 1/2T, T - ID: IT-T-001					EN ISO 148-1	-

Individual Values	# 1	# 2	# 3
Value [J]	36	30	37
Statistics	Mean	Min/Max	Std Dev
	34.33	22	

Individual Values	# 1	# 2	# 3
Value [J]	62	65	58

Statistics	Mean	Min/Max	Std Dev
	61.67	38	

Surface hardness162HB180HB-

Physical Properties

Property	Symbol	Actual	Target/Min	Maximum	Method	Status
Grain size		6	4	9	ASTM E112	-
Purity K4(ox)		20	-	35	DIN 50602	-
Purity K4(s)		25	-	35	DIN 50602	-
Straightness		1.2mm/m	-	2.0mm/m		-
Reduction ratio		3.8	3.0	-		-
Residual magnetism		5A/cm	-	10A/cm		-
fully formed structure, - no residue of dendritic - cast structure		Yes	-	-		-

Supplementary Tests

Property	Actual	Target/Min	Maximum	Method	Status
Ultrasonic test 100%	Yes Pass	-	-	EN 10228-3 class 3	-
Surface test	Yes 100% visually inspected, surface free from defects	-	-		-
Radioactivity	Yes max 100 Bq/kg Bq/kg	-	-		-

Validation

We hereby certify that the material described above has been manufactured, tested and inspected in accordance with the requirements of the applicable specifications and standards, and that the results comply with the specified requirements.

Individual Statements

✔ Material is of non-Russian origin (EU Regulation No. 833/2014)

Validated By

Name	Title	Department	Date
Dr. Maria Fischer	Head of Quality Assurance	Quality Assurance	2025-04-10